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Jiawei Dachaihu decoction protects against mitochondrial dysfunction in atherosclerosis (AS) mice with chronic unpredictable mild stress (CUMS) via SIRT1/PGC-1α/TFAM/LON signaling pathway

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ABSTRACT

Ethnopharmacological relevance: JiaWei DaChaiHu is composed of *Bupleurum chinense*, *Scutellaria baicalensis*, *Pinellia ternata*, *Paeonia lactiflora*, *Zingiber officinale* Roscoe, *Poncirus tuifoliata*, *Rheum palmatum* L., *Curcumae Radix*, *Herba Lysimachiae*, *Ziziphus*. JiaWei DaChaiHu is one of the most common traditional Chinese medicines for the treatment of depression.

Aim of the study: The chronic unpredictable mild stress (CUMS) has been shown to promote atherosclerosis (AS). Dachaihu has been widely used in traditional Chinese medicine and has been known to exert distinct pharmacological effects. This investigation aims to examine the therapeutic effect of Jiawei Dachaihu extract on AS animal models with CUMS.

Methods: AS-CUMS mice model was established by Apoe$^{-/-}$ mice. Mice were treated with Jiawei Dachaihu. Serum total cholesterol (TC), triglyceride (TG), low-density lipoprotein (LDL-C), high-density lipoprotein (HDL-C) levels were measured using ELISA kits. Aortic tissue pathologic changes detected by oil red O staining. Mice behavioral changes detected by sucrose preference test and sucrose preference test. The relative mRNA expression levels of CRH, ND1, and TFAM were determined by qRT-PCR. 5-HT1A, BDNF, LON, TFAM, PGC-1α, and SIRT1 protein expression determined by western blotting. ATP content detected by ATP kits.

Results: The treatment with Jiawei Dachaihu extract alleviated the veins plaque and reduced stress signs *in vitro* and *in vivo*. It increased the ATP and HDL-C levels while decreased the TC, TG, LDL-C levels. Jiawei Dachaihu extract treatment upregulated Lon, SIRT1, TFAM, PGC-1α, BDNF, and 5-HT1A protein expression and regained mitochondrial function.

Conclusion: Jiawei Dachaihu extract could alleviate AS and reduce CUMS by upregulating the SIRT1/PGC-1α signaling and promoted its crosstalk with Lon protein to maintain mitochondrial stability.

1. Introduction

People with depression are at greatly increased risk for obesity, metabolic syndrome, coronary artery disease, and inflammation (Lin et al., 2023). Depression and high-fat diet are renowned independent risk factors for atherosclerosis and other cardiovascular disorders, propound the interaction of physiological and psychological factors in the progression of these diseases (Li et al., 2019; Wang et al., 2014). Liver is the body's largest detoxification organ, if the patient liver lesions, such as hepatitis, mostly accompanied by anorexia, nausea, sclera or skin yellowing and other symptoms, can affect the normal function of the liver, easy to interfere with the liver detoxification effect, resulting in excessive accumulation of toxins in the body and can not be discharged, and then blocked the operation of the qi and blood, you will produce emotional depression symptoms, may also appear delirious performance, and in severe cases may even depression. The theory of Traditional Chinese Medicine posits that patients with liver disease are generally more prone to depression, mainly because liver damage can block the flow of qi and blood in the body, producing symptoms of

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Abbreviation

CUMS	chronic unpredictable mild stress
AS	atherosclerosis
SIRT1	Sirtuin 1
PGC-1α	PPARG coactivator 1 alpha
TFAM	transcription factor A
LON	Lon protease
TC	total serum cholesterol
TG	triglyceride
LDL-C	low-density lipoprotein
HDL-C	high-density lipoprotein
CRH	Corticotropin releasing hormone
ND1	NADH dehydrogenase subunit 1
5-HT1A	5-hydroxytryptamine receptor 1A
BDNF	brain derived neurotrophic factor
ApoE	apolipoprotein E
NAFLD	nonalcoholic fatty liver disease
mtDNA	Mitochondrial genome
ATP	Adenosine triphosphate
ROS	reactive oxygen species
HFD	high fat diet
ELISA	Enzyme-Linked Immunosorbent Assay
qRT-PCR	Quantitative real time polymerase chain reaction
PMA	phorbol-12-myristate-13-acetate

depression, and depression impacts hepatic function, thereby implying a connection between depression and lipid metabolic disorders (Wang et al., 2014). Accumulating evidence indicated that chronic stress is a critical risk factor for atherosclerosis (Empana et al., 2005), and the research on the underlying mechanism by which chronic stress factors engaged in the development of atherosclerosis is increasing (Everson-Rose et al., 2014; Peplinski et al., 2018). Previous studies have reported that chronic psychological stress induces chronic inflammation in the aortic wall, thereby promoting atherosclerosis in animal models (Bernberg et al., 2012). Additionally, chronic stress has been found to downregulate PPARγ/LXRα/ABCA1 via HMGB1/TLR4-mediated pathways, leading to the development of atherosclerosis in ApoE$^{-/-}$ Mice (Gu et al., 2019).

There is a growing body of evidence suggesting that chronic stress plays a significant role in the development of atherosclerosis (Yang et al., 2019). In Chinese medicine, depression is categorized as a "depression syndrome", which is an imbalance of emotions caused by excessive worry and accumulation of anger. As for the complex etiology of depression, some scholars believe that the root cause of its development lies in the blockage of qi (Chen and Guo, 2017). Although Western medicine has achieved some success in the treatment of depression, there are some potential risks such as adverse drug reactions and easy relapse after stopping the medication when taking it for a long period of time. In recent years, traditional chinese medicine based on the whole, identification and treatment and treatment of small side effects and other characteristics, showing obvious therapeutic advantages (Ruan et al., 2023). According to Chinese medicine, dredging the liver and regulating qi, nourishing the heart and calming the mind are fundamental to the treatment of depression. Dachaihu is able to regulate anxiety and depression, as well as anti-inflammatory and regulating the function of the gastrointestinal tract. Dachaihu decoction has been found to alleviate high fat diet-induced nonalcoholic fatty liver disease (NAFLD) through remodeling the gut microbiota and modulating the serum metabolism (Cui et al., 2020). Dachai hu extract can elevate the levels of lactic acid, blood sugar, and saturated fatty acid, and reduced levels of high-density lipoprotein and unsaturated fatty acid (Law et al., 2014), also increase plasma levels of serotonin (5-HT) (Kotoulas et al., 2006). As autophagy regulates lipid and glucose metabolism, loss of autophagy can cause abnormal accumulation of lipids and significantly increase plasma triglycerides, with reductions in fatty acid beta-oxidation (Singh et al., 2009).

Mitochondria is responsible for various vital cellular processes, like apoptosis, redox signaling, maintenance of calcium homeostasis, and lipid metabolism (Raghu et al., 2021), and it plays a critical role in maintaining energy homeostasis and coping with cellular stress (Roca-Agujetas et al., 2019). The PGC-1α and SIRT1 control energy expenditure and are vital links in the regulatory network for metabolic homeostasis. Furthermore, they yield advantageous outcomes in terms of physical activity and dietary intervention (Cantó and Auwerx, 2009). Besides, PGC-1α and SIRT1 present inside mitochondria and are close to mtDNA, and interact with mitochondrial transcription factor A (TFAM). The mitochondria PGC-1α and SIRT1 are nuclear counterparts that mediate the crosstalk between nuclear and mitochondrial genome and mitochondrial biogenesis (Aquilano et al., 2010). On the other hand, the Lon protein is the central protease in the mitochondrial matrix in eukaryotes; it regulates mitochondrial transcription by stabilizing the mitochondrial TFAM: mtDNA ratio via selective degradation of TFAM (Matsushima et al., 2010). These data suggest crosstalk between Lon and PGC-1α and SIRT1 axis.

Further investigation is necessary to establish a comprehensive foundation regarding the Jiawei Dachai hu extract and its potential as a therapeutic intervention for chronic stress disorders. The CUMS method, also known as the Chronic Unpredictable Moderate Stress method, is close to the onset of human depression through environmental factors that induce depression. In order to gain a deeper understanding of the impact of Jiawei Dachai hu on individuals with AS-CUMS, the current study implemented a treatment regimen using Jiawei Dachai hu on an AS-CUMS mouse model.

2. Materials and methods

2.1. Animal experiments

2.1.1. Reagents

The human monocytic cell line THP-1 was purchased from [Gefan Bio, Shanghai]; ATP detection kit [Jiancheng Bio, Nanjing]. ROS detection kit [Sigma, Massachusetts]. Antibodies [5-HT1A, BDNF, Lon, SIRT1, PGC-1α, TFAM, and GAPDH] were purchased from [Invitrogen, California]. Oil red O staining dye was purchased from [Sigma, Massachusetts]. Ox-LDL was purchased from [Sigma, Massachusetts]. Resveratrol was purchased from [InvivoChem, USA]. Cortisol was purchased from [Sigma, Massachusetts]. Simvastatin was purchased from [Sigma, Massachusetts]. St John's herbal extract was purchased from [Gefan Bio, Shanghai]. JiaWei DaChaiHu extract was purchased from [Gefan Bio, Shanghai]. JiaWei DaChaiHu composition: *Bupleurum chinense* DC. 9g, *Scutellaria baicalensis* Georgi 9g, *Pinellia ternata* (Thunb.) Makino 9g, *Paeonia lactiflora* Pall. 15g, *Zingiber officinale* Roscoe 3 slices, *Citrus* × *aurantium* L. 6g, *Rheum palmatum* L. 6g, *Curcuma aromatica* Salisb. 30g, *Ouret lanata* (L.) Kuntze 30g, *Ziziphus jujuba* Mill. 5g. All herbs were purchased from the Gefan Bio (Shanghai, China). According to the TCM decoction method, all herbs were mixed and boiled for 1.5 h, and the process was repeated three times. Chemical fingerprint and main chemical components of JiaWei DaChaiHu were shown in Figure S1, Figure S2, Table S1 and Table S2.

2.1.2. Preparation of medicated serum

The experimental animals were given flavored Dachaihu extract twice a day for seven days, and with regard to the equivalent dose for humans and animals, the equivalent dose for mice was equivalent to 9.1 times that for humans, on the basis of dose per unit of body weight. The blood samples were withdrawn from the abdominal aorta on day eight after administration, and serum is separated by centrifugation at 3000 rpm/min for 10 min at 4 °C. The serum was then collected and deactivated for 30 min at 56 °C and followed by filtration using a Micropore filter. The final collected serum was stored at −20 °C for further experiment. The serum was used within one month of preparation.

2.1.3. Animals grouping and drug administration

The AS-CUMS animal model was prepared by using $ApoE^{-/-}$ mice. Animals were banned from water access one night before experiments. Firstly, the mice were set to a high-fat diet (standard food plus 2% cholesterol, 10% pig oil, and 0.5% sodium bilate) for one week to establish AS model. 12 h later, blood samples were withdrawn from the orbit vein and collected by centrifugation for further analysis. The AS-CUMS model was prepared by utilizing the previously established AS model and stressed using various mild stress sources as shown in Table 1 for 8 weeks. Mice body weight was measured weekly, and their food consumption, water drinking, activity, excitement, emotional response, sleep rate, hair status, mucosa status, and ear pigmentation were observed and recorded.

C57BL/6 mice and $ApoE^{-/-}$ mice (Male, 5 weeks, 17.0 ± 0.5 g) were divided into 5 groups (N = 6): Control group (C57BL/6 mice), $Apoe^{-/-}$/HFD group ($ApoE^{-/-}$ mice with high fat diet), $ApoE^{-/-}$/HFD + CUMS group, $Apoe^{-/-}$/HFD + CUMS + Simvastatin + St. John's wort extract group (Simvastatin for lowering blood lipids, St. John's wort extract for antidepressant), $ApoE^{-/-}$/HFD + CUMS + Jiawei Dachaihu (mice were treated by Jiawei Dachaihu decoction). The blood samples were withdrawn from the abdominal aorta on day eight after administration, and serum is separated by centrifugation at 3000 rpm/min for 10 min at 4 °C. The serum was then collected and deactivated for 30 min at 56 °C and followed by filtration using a Micropore filter. The final collected serum was stored at −20 °C for further experiment. The serum was used within one month of preparation.

2.1.4. ATP assay

The cells were harvested (1×10^6/group), and 300–500 μL of hot double-distilled water was added to the cells and place in a hot bath (90–100 °C) to absorb plasma breakage, then, the cell suspension was heated in a boiling water bath for 10 min. The mixture was then transferred and tested for ATP content using ATP content Kit following the manufacturer's protocol.

2.1.5. ELISA assay

The total serum cholesterol (TC), triglyceride (TG), low-density lipoprotein (LDL-C), high-density lipoprotein (HDL-C) levels were measured using ELISA kits [Invitrogen, California] following the manufacturer's protocol.

Table 1
Schedule for CUMS.

Sr. No	Day	Type of stress procedure	Time (h)
1	Monday	Food and water deprivation	24 (9 a.m.–9 am)
2	Tuesday	Empty bottle; Foreign object, (i.e. marbles)	1 (9 a.m.–10 a.m.); 24 (9 a.m.–9 am)
3	Wednesday	Overnight illumination; forced swimming	10 (8 p.m.-6 am); 1 (4 min/animal) (9 a.m.–10 a.m.)
4	Thursday	Cage tilt; restraint	5 (12 p.m.–5 pm); 1 (10 a.m.–11 a.m.)
5	Friday	Food deprivation; Soiled cage (with water)	12 (9 a.m.-9 pm); 12 (9 p.m.-9 am)
6	Saturday	Water deprivation; overnight illumination	12 (10 a.m.-10 p.m.); 12 (7 p.m.-7 am)
7	Sunday	Empty bottles; Cage tilt	2 (10 a.m.–12 a.m.); 5 (12 a.m.-5 pm)

2.1.6. Oil red O staining

Aortic tissue specimens were sectioned at a 10-μm diameter. The aortic sinus and aortic valve transverse sections were stained with Oil Red O and counterstained with Gill III hematoxylin.

2.2. Behavior test

2.2.1. Sucrose preference test

The experiment was divided into a training period and a test period. After the mice have completed the adaptive breeding, they received double bottle training. First, we gave the mice two bottles of 1% (w/v) sucrose water for 24 h, and then we gave them one bottle of 1% (w/v) sucrose water and pure water for 24 h at the same time. During the test period, the mice were fasted for 23 h before the test, then the sugar water consumption test was carried out for 1 h (3:00–4:00 pm), as each mouse was given 100 mL 1% (w/v) sucrose water and pure water at the same time. After drinking for 1 h, to prevent the mice from habitually drinking the water on one side, the two water bottles were switched every half an hour. The water and sucrose consumption were recorded, and the percentage of sugar water consumption was calculated. The calculation formula was: sugar water partiality = sugar water consumption/total fluid intake × 100%.

2.2.2. Open field test

Each mouse was placed in the center of a self-made open box consisting of a 20 × 20 cm^2 at the mouse motion score was observed and recorded within 3 min (1) Horizontal score: The number of times mice crossed the ground grid, and the animal crossed one box (3 claws and above) was 1 point; 1 min per 10 cm walking along with the animal; (2) Vertical score: This is the number of times the mouse is upright (two forefingers vacate off the ground or climb to the open box). The total score for the field experiment = Horizontal Score + Vertical Score. After each test, we avoided odor interference on the mice by thoroughly cleaned the field with 75% alcohol.

2.3. Transmission electron microscope observation of mitochondrial morphology

Specimens of previously extracted aortic tissues were fixed in 2.5% glutaraldehyde for 2 h, washed in 0.1 M phosphate-buffered saline (pH 7.4), then processed by post-fixation in 1% osmium tetroxide for 2 h. After dehydration in graded alcohols, propylene oxide replacement, impregnation, and embedding with Spurr resin, and polymerization at 70 °C, the tissues were cut into 70-nm ultrathin sections using an ultrathin slice. Next, we selected random specimens for analysis, stained them with lead citrate and uranyl acetate, and analyzed them by TEM (Hitachi TEM systems, Tokyo, Japan).

2.4. qRT-PCR

The relative mRNA expressions of CRH, ND1, and TFAM were determined by qRT-PCR. The total RNA in the aortic tissues of each mice was extracted using an RNA extraction kit following the manufacturer's protocol. The RNA concentration was measured by spectrophotometer, and cDNA synthesis was done using cDNA reverse transcription Kit. The qRT-PCR protocol was done as previously reported [36]. The primer sequences of each gene were as following: siRNA-TFAM-316 (5′-3′ Forward: GUGGCAGGUAUAUAAAGAATT, 5′-3′ Reverse: UUCUUUAUAUACCUGCCACTT). siRNA-TFAM-474 (5′-3′ Forward: GUUCAGCUUAUAACGUUUATT, 5′-3′ Reverse: UAAACGUUAUAGCUGAACTT). siRNA-TFAM-693 (5′-3′ Forward: GUCGCACAAUAAAGAAACATT, 5′-3′ Reverse: UGUUUCUUUAUUGUGCGACTT).

2.5. Western blotting

Western blotting analysis was conducted to measure 5-HT1A, BDNF, LON, TFAM, PGC-1α, and SIRT01 protein expression. The total protein from cells or tissue samples were extracted using a radioimmunoprecipitation assay buffer [Invitrogen, California]. An equal amount of protein was loaded onto SDS-PAGE, and the protein expressions were analyzed as previously reported (Dong et al., 2019).

2.6. Cell experiments

2.6.1. Cell culture and cell transfection

THP-1 cells were incubated in RPM1640 medium and supplemented with 160 nmol/l phorbol-12-myristate-13-acetate (PMA), 10% of fetal bovine serum. At 80% confluency, the cells changed from suspended to adherent. Upon cellular morphology changed to a spindle, oval, or irregular shape, it indicates that the monocytes differentiate into macrophages, then the cells were counted to form the macrophage test group cells. For cell transfection, the cells were transfected with Lipofectamine 3000 (Invitrogen, USA) following the manufacturer's protocol (Zou et al., 2020).

2.6.2. Foaming cell modeling

The THP-1 derived macrophages were generated by culturing the cells in 0.3% VSA's serum-free RPMI1640 culture medium and the addition of 50 mg/L ox-LDL. Then the cells were incubated at 37 °C in a 5% CO_2 incubator for 48 h until the macrophage-derived foam cells were formed.

2.6.3. MTT assay

Cell viability was evaluated using MTT assay. The THP-1 cells were incubated accordingly, and following each experimental group design for the indicated period, the cell viability was measured as previously reported (Fu et al., 2019).

2.6.4. ROS assay

1×10^4 cells were cultured in laser focusing-specific cell culture dishes (35 mm) for 24 h. Then the cells were treated with either SIRT1 inhibitor/activator and followed by the addition of MitoSOX for 15 min. The mitochondrial ROS-specific fluorescent was measured.

2.7. Statistical analysis

Data were presented as means ± SEM. One-way ANOVA or Student's t-test was used to determine the statistical significance, and $P < 0.05$ was considered significant.

3. Results

3.1. Jiawei Dachaihu decoction alleviated atherosclerosis plaque and serum lipid levels in vivo

Jiawei Dachaihu decoction effect on atherosclerosis animal model with CUMS were examined. Compared with the animals in ApoE$^{-/-}$/HFD and ApoE$^{-/-}$/HFD + CUMS groups, Jiawei Dachaihu decoction treatment increased the ATP and HDL-C levels (Fig. 1A and 1D). On the other hand, Jiawei Dachaihu decoction treatment significantly decreased the TC, TG and LDL-C levels compared with the mice in ApoE$^{-/-}$/HFD + CUMS groups (Fig. 1B, 1C, 1E). Moreover, Oil red O staining results showed that, Jiawei Dachaihu decoction treatment significantly reduced the lesions areas in the mice's aortic sinus and aortic valves (Fig. 1F).

3.2. Jiawei Dachaihu decoction reduced stress signs in ApoE$^{-/-}$ mice

Stress signs in the mice treated with Jiawei Dachaihu decoction were slightly reduced compared to the mice in the ApoE$^{-/-}$/HFD + CUMS group. The Jiawei Dachaihu decoction treatment affected sucrose

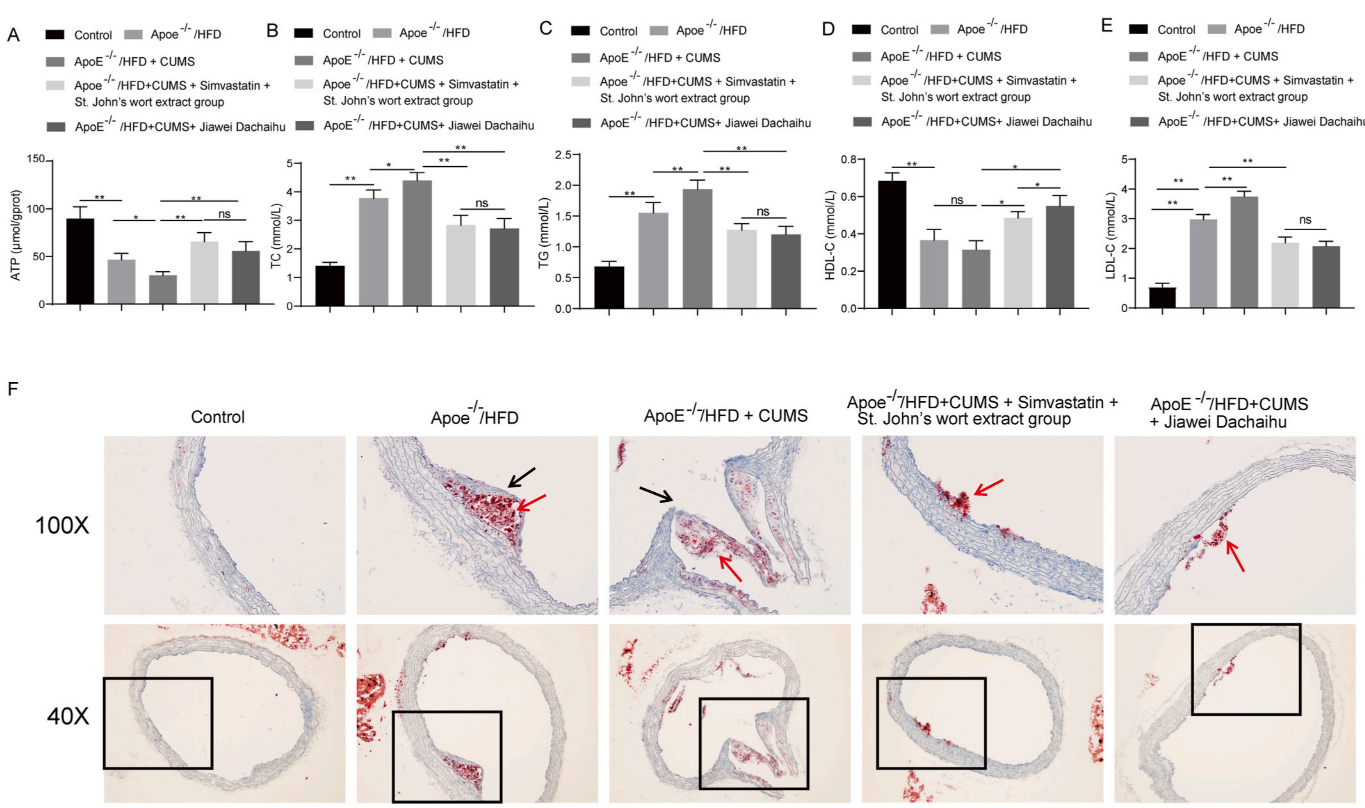

Fig. 1. The effect of Jiawei Dachaihu on veins plaque and the ATP, HDL-C, TG, TC, and LDL-C levels. (A) Detection of ATP levels by ATP assay. **(B-E)** Detection of TC, TG, HDL-C, LDL-C levels in mice serum by biochemical analyzer. **(F)** Atherosclerotic lesions in aortic sinus detected by Oil red O staining.

preferences (Fig. 2A). Meanwhile, average movement speed, duration of movement of the open field all around significantly increased (Fig. 2B–2E). These results suggested the potential role of Jiawei Dachaihu decoction in reducing stress signs in AS-CUMS animal models.

3.3. Jiawei Dachaihu decoction protected mitochondrial by modulating the key elements of mitochondrial biogenesis and function

Compared to ApoE$^{-/-}$/HFD + CUMS group, Jiawei Dachaihu decoction treatment significantly reduced the CRH mRNA expression level (Fig. 3A). Also, it increased the protein expression level of the 5-HT1A and BDNF protein which would improve the mitochondrial morphology and function (Fig. 3B and 3C). Western blotting analysis shown that Jiawei Dachaihu decoction treatment promoted the protein expression level of Lon, TFAM, PCG-1α, and SIRT1 (Fig. 3D).

3.4. SIRT1 is vital for mitochondrial DNA transcription

Macrophage cells (THP1 cell) were isolated to determine the vitality of SIRT1 in macrophage formation. Oil red O staining results shown that the macrophage successfully foamed (Fig. 4A). After treated with SIRT1 activator (resveratrol), the expression of SIRT1, PGC-1α, TFAM, and Lon proteins increased, compared with mitochondrial stress treatments (ox-LDL group and Cortisol group) (Fig. 4B). Besides, resveratrol significantly increased the ATP levels and promoted the gene expression of ND-1 compared to the ox-LDL and Cortisol groups (Fig. 4C and 4D). These findings indicated that SIRT1 is critical for mitochondria function and DNA transcription.

3.5. The Lon protein interoperated with the TFAM protein

Immunoprecipitation assay was utilized to assess the possible crosstalk between Lon and TFAM proteins (Fig. 5A). Multiple siRNAs-TFAM have been constructed to knock down TFAM protein expression, and siRNA-693 had been shown to inhibit TFAM gene expression significantly (Fig. 5B). Then, we collected the serum from the mice treated with Jiawei Dachaihu decoction to determine the effect of silencing the TFAM gene on the protective effect of Jiawei Dachaihu decoction. As seen in Fig. 5C, this was the added values of the serum containing Jiawei Dachaihu decoction. In Fig. 5D, the expression of TFAM was significantly decreased in the model group than in the control group. Also, the addition of the serum containing Jiawei Dachaihu decoction increased the TFAM expression and slightly reduced Lon expression compared to the model group, and no changes were found in the protein expression of PGC-1α and SIRT1. On the other hand, silencing TFAM in the presence of the serum containing Jiawei Dachaihu decoction has slightly reduced the TFAM expression and slightly increased Lon expression while no changes were found in the protein expression of PGC-1α and SIRT1.

3.6. Knockdown of TFAM has reduced ATP levels, interrupted mtDNA transcription, and promoted ROS production

The ATP levels reduced in the TFAM silenced group compared to the group contained Jiawei Dachaihu decoction (Fig. 6A). Also, TFAM silencing significantly reduced the mRNA expression level of ND-1 (Fig. 6B) and promoted ROS production compared to the group contained Jiawei Dachaihu decoction (Fig. 6C).

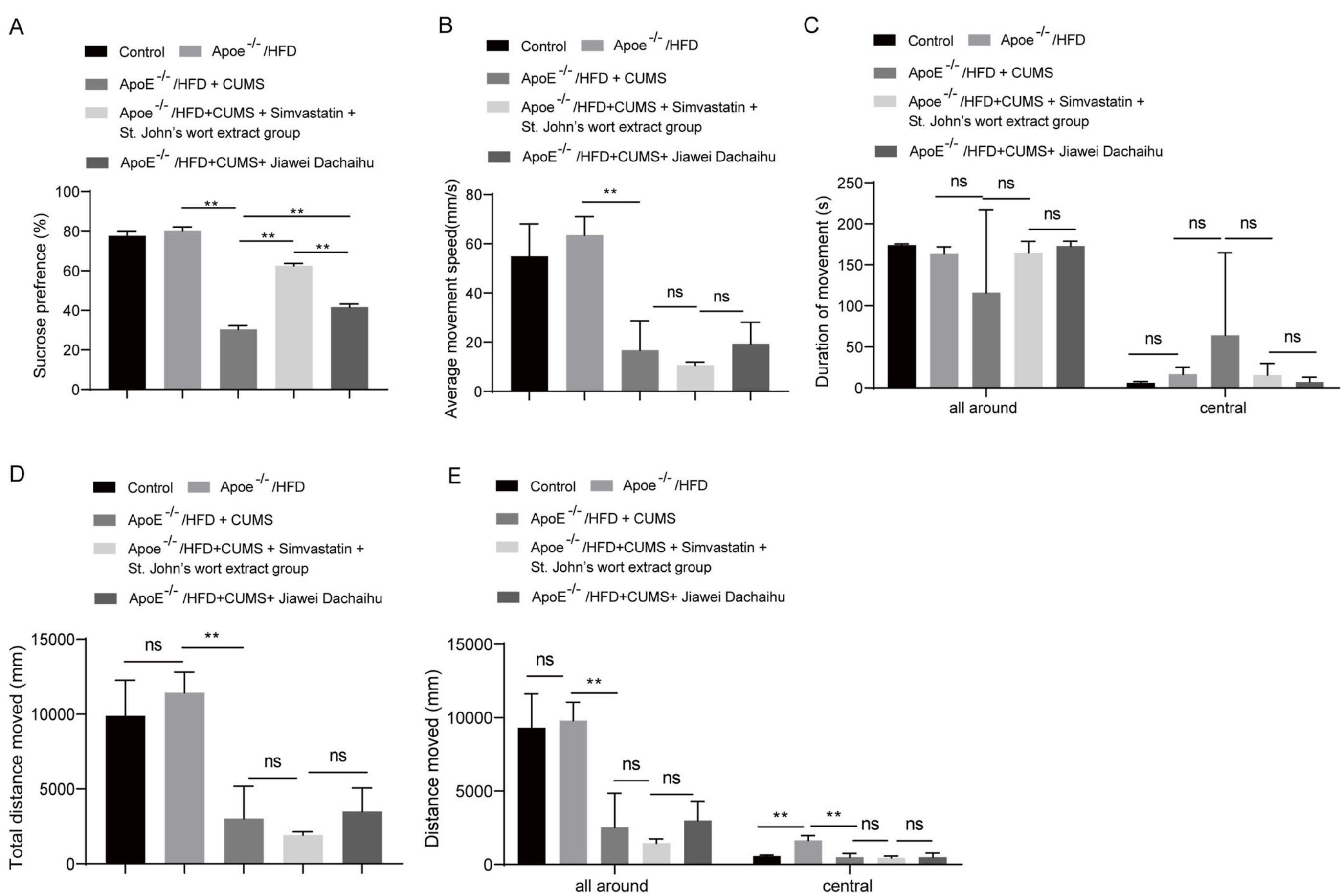

Fig. 2. Jiawei Dachaihu decoction treatment has significantly reduced the stress signs in the experimental animals. (A) The sucrose preference. **(B-E)** The average movement, duration movement, total moved distance and moved distance of the open field analysis.

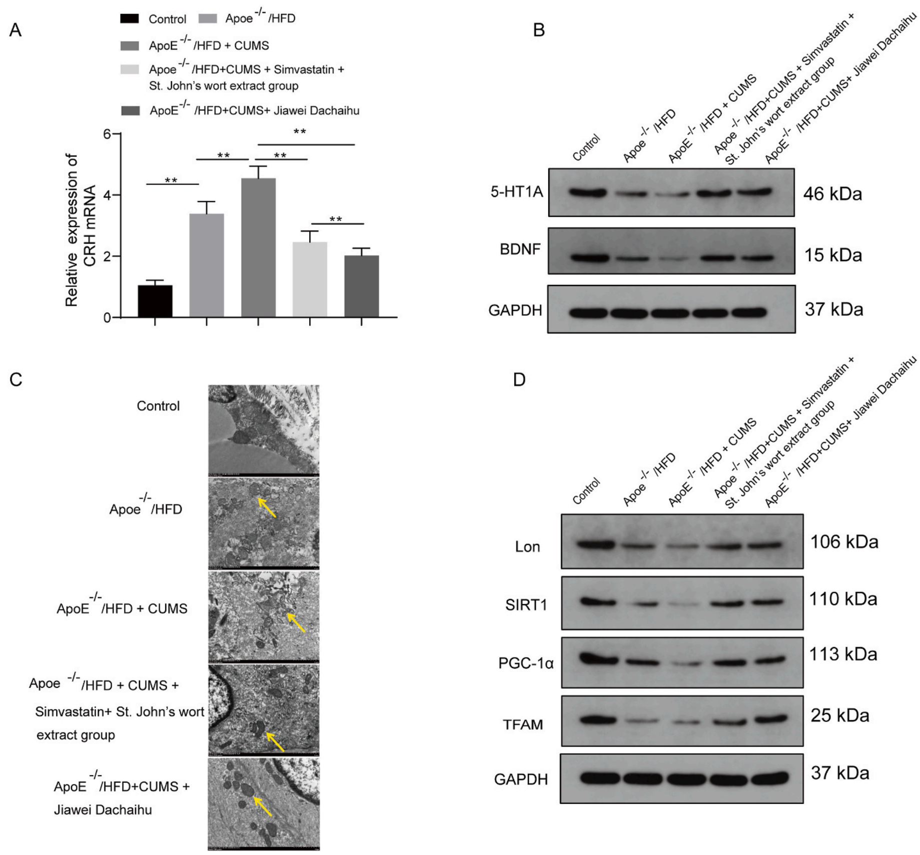

Fig. 3. Jiawei Dachaihu decoction protected mitochondrial by modulating the key elements of mitochondrial biogenesis and function (A) qPCR analysis of the CRH gene expression after the treatment with Jiawei Dachaihu in animal models. **(B)** Western blotting analysis of the 5HT1A and BDNF protein expression. **(C)** ECM analysis of the mitochondrial structure. **(D)** Western blotting analysis of Lon, SIRT1, PGC-1α, and TFAM protein expression.

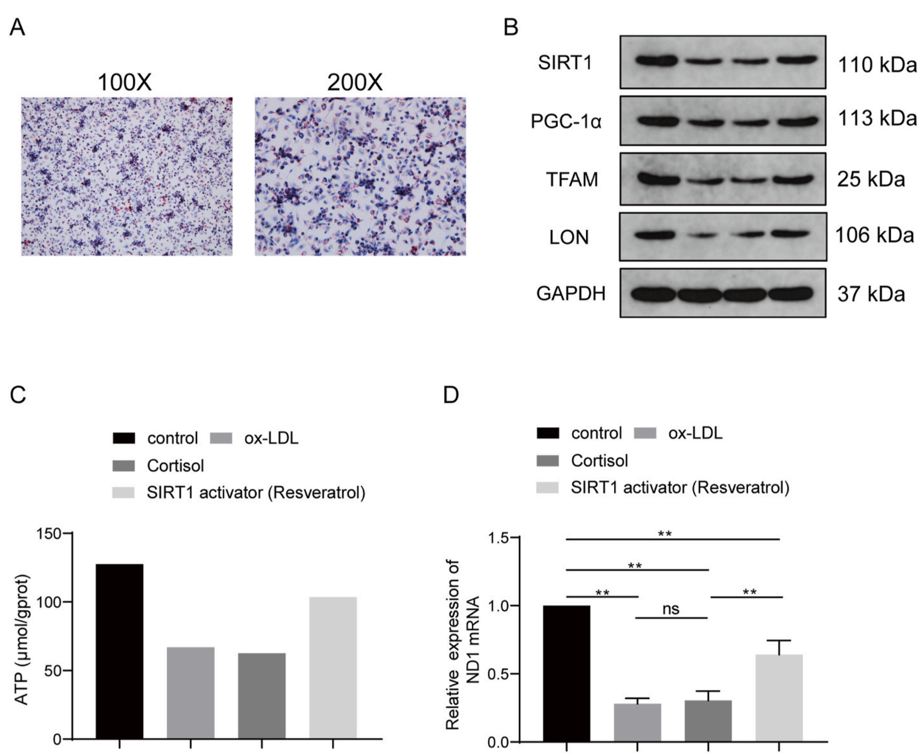

Fig. 4. SIRT1 is vital for mitochondrial DNA transcription. (A) Foam cell formation was identified using Oil red O staining in the THP1 cell. **(B)** Western blotting analysis of SIRT1, PGC-1α, TFAM and Lon protein expression level. **(C)** ATP levels measured by ATP assay. **(D)** ND1 mRNA expression detected by qRT-PCR.

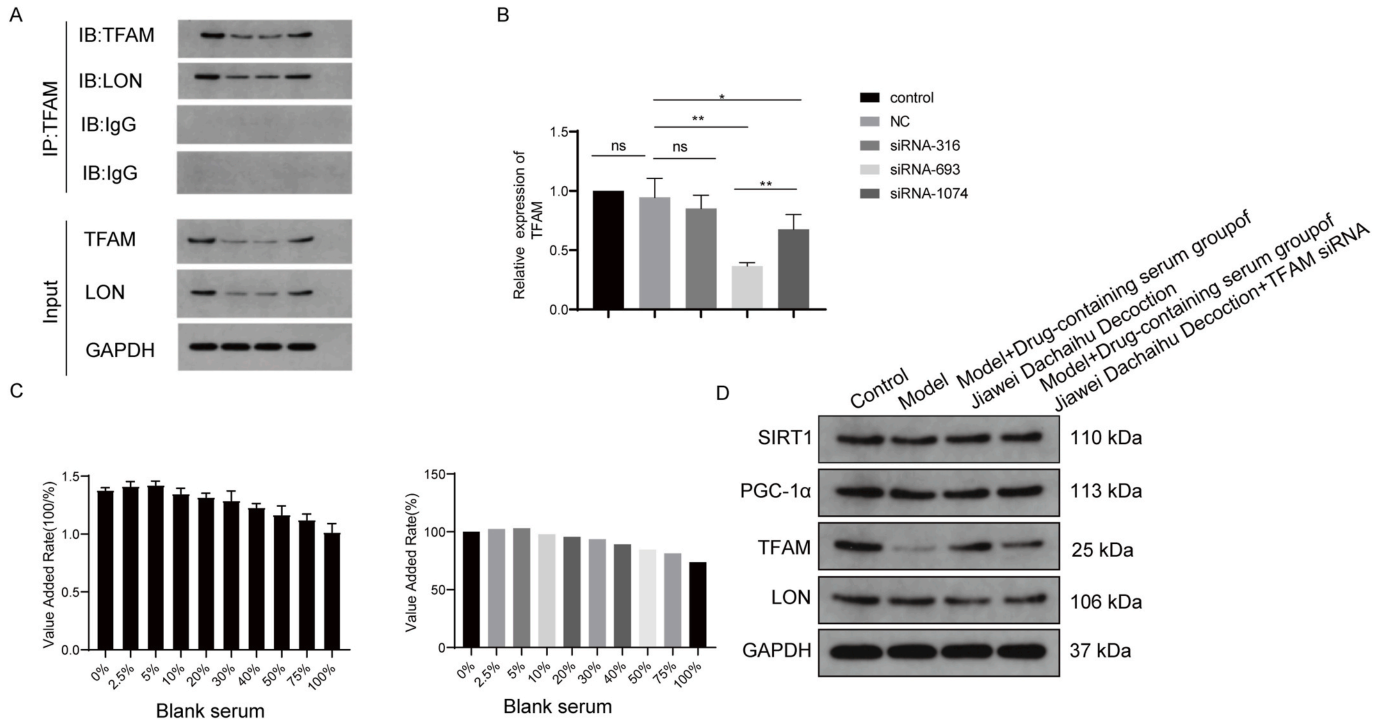

Fig. 5. Lon and TFAM interacted and were involved in the mitochondrial functions. **(A)** Immunoprecipitation assay determined the interaction between Lon and TFAM proteins. **(B)** TFAM mRNA expression level detected by qRT-PCR. **(C)** The administered levels of the medicated blood serum from the animal models. **(D)** Western blotting analysis of SIRT1, PGC-1α, Lon, and TFAM protein expression.

4. Discussion

There is a growing body of evidence that suggests a strong association between stress-induced hyperlipidemia and heightened oxidative stress with AS. (Amor et al., 2017; Devaki et al., 2013). In a prior investigation, rats subjected to chronic stress exhibited elevated serum levels of total triglycerides, cholesterol, low-density lipoprotein cholesterol, and very low-density lipoprotein cholesterol, while the concentration of high-density lipoprotein cholesterol remained unaltered when compared to control rats (Meng et al., 2018). Dachaihu has been seen to be efficient in the treatment of rib-side fullness, fever, chills, abdominal pain, and fullness (Li et al., 2014). Also, it has been reported to treat non-alcoholic fatty liver disease (Yang et al., 2019) and exhibit antihypercholesterolemic effects (Yoshie et al., 2004). JiaWei DaChaiHu includes *Bupleurum chinense* DC., *Scutellaria baicalensis* Georgi, *Pinellia ternata* (Thunb.) Makino 9g, *Paeonia lactiflora* Pall., *Zingiber officinale* Roscoe, *Citrus* × *aurantium* L., *Rheum palmatum* L., *Curcuma aromatica* Salisb., *Ouret lanata* (L.) Kuntze, *Ziziphus jujuba* Mill. The main components of *Scutellaria baicalensis* Georgi include polysaccharides, flavonoids, flavonoid glycosides, volatile oils, and pharmacological studies have shown that it has antibacterial, anti-inflammatory and antipyretic, hepatoprotective, choleretic, sedative and other effects, and baicalein in its composition can inhibit collagen arachidonic acid and so on, so as to achieve the purpose of inhibiting blood lipids and improving the aggregation of platelets. Pinellia ternata (Thunb.) Makino is composed of alkaloids, dihydroxybenzaldehyde glucoside and other main components, its effects include suppressing cough, antiemetic, inhibiting gastric acid secretion, etc., due to the effect of drying dampness and resolving phlegm, lowering reversal and stopping vomiting, it also has a certain effect on the regulation of blood lipids. Ginger alcohol extract in *Zingiber officinale* Roscoe has the effect of lowering serum cholesterol, on the basis of improving the lipid metabolism of liver, it can enhance the lipid metabolism of the body and achieve the effect of lowering blood lipids. The pharmacology of Ziziphus jujuba Mill shows that the main components include protein, sugar, vitamins, etc. Cyclic adenosine monophosphate has a certain effect on the improvement of blood lipids, and due to the richness of vitamins, it helps to regulate the absorption of nutrients by adding jujube to enhance the effect of blood lipid regulation. Citrus × aurantium L. is mainly composed of hesperidin, octreotide, vitamins, etc. It has the effect of improving arterial blood flow, regulating gastrointestinal absorption, etc., which can enhance the therapeutic effect of Chaihu in the treatment of hyperlipidemia. Rhubarb acid in rhubarb is the main component of laxative effect, which can promote gastrointestinal absorption, increase peristalsis, promote defecation, and promote fat metabolism in the treatment of hyperlipidemia. The findings of this study demonstrate that Jiawei Dachaihu decoction effectively enhances ATP and HDL-C levels, while reducing TG, TC, and LDL-C levels. Additionally, it mitigates aortic sinus and valve lesions. Furthermore, treatment with Jiawei Dachaihu decoction significantly alleviates stress-related symptoms in the experimental animal models.

According to previous reports, significant proteins play a crucial role in the regulation of mitochondrial function and biogenesis. The current study demonstrates that the administration of Jiawei Dachaihu resulted in an increase in the protein levels of 5-HT1A and BDNF, while simultaneously decreasing the mRNA level of CRH. It is worth noting that CRH has been found to influence mitochondrial morphology and function (Battaglia et al., 2020). 5-HT acts through the 5-HT1A receptor subtype and promotes axonal transport of mitochondria (Chen et al., 2008), and there are significant correlations between BDNF, antioxidant enzyme activities, and mtDNAcn(Wang et al., 2020). Moreover, The Lon-TFAM and PCG-1α/SIRT1 axes are essential in the regulation of mitochondrial functions and mtDNA transcription, suggesting crosstalk between the two axes (Araujo et al., 2018; Spiegelman, 2007; Voos and Pollecker, 2020). It has been known that ox-LDL and cortisol induce mitochondrial stress and inhibit its function (Manoli et al., 2007). SIRT1 is a vital regulator of the mtDNA, and one of its transcripts is ND1 (Li et al., 2016) and that resveratrol induces the expression and activity of SIRT1 (Chao et al., 2017). As a key factor in mitochondrial bioactivity and energy metabolism, PGC-1α is widely involved in maintaining the

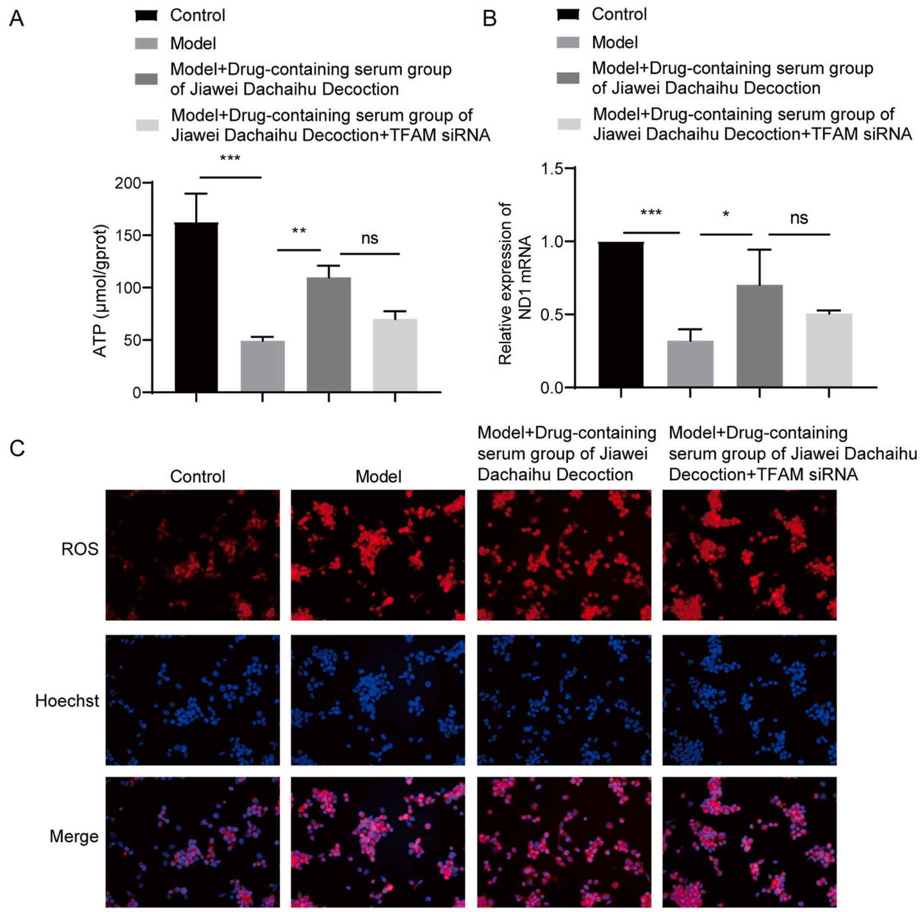

Fig. 6. TFAM was vital for the protective effect of Jiawei Dachaihu decoction. (A) The ATP levels measured by ATP assay. **(B)** ND1 mRNA expression detected by qRT-PCR. **(C)** The ROS levels measured by ROS detection kit.

normal function of the cardiovascular system, and its dysfunction is thought to be associated with numerous cardiovascular diseases such as atherosclerosis, heart failure, and aortic coarctation (Wei et al., 2021). In this research, Jiawei Dachaihu decoction treatment upregulated the PGC-1α/SIRT1 signaling pathway and its subsequent signaling proteins, Lon and TFAM. TFAM expression lreduction eads to abnormal mitochondrial DNA packaging and inflammation due to released mitochondrial DNA, which may exacerbate plaque buildup in atherosclerosis (Cobo et al., 2022).

The process of mitochondrial lipid metabolism encompasses the synthesis of cardiolipin, phospholipids such as phosphatidylethanolamine and phosphatidylglycerol. The latter serves as a precursor for the production of late endosomal lipid phosphate. Additionally, mitochondrial fatty acid synthesis is involved in this process. A developing hypothesis posits that psychosocial and behavioral factors can be detected, integrated, and transmitted by the mitochondria, resulting in cellular and molecular alterations. Furthermore, mitochondrial signaling plays a role in the biological incorporation of psychological states (Picard and McEwen, 2018). Since many enzymes and transport processes are parts of the mitochondrial lipid metabolism, lipid-related disorders could be increasingly expected (Mayr, 2015). Jiawei Dachaihu improved the mitochondrial morphology and function, which implied it may have protective effect for mitochondrial lipid metabolism.

The current study demonstrated the interrelationship between Lon and TFAM proteins and their involvement in maintaining mitochondrial stability. Additionally, the importance of TFAM in the protective effects of Jiawei Dachaihu decoction was highlighted, as it upregulated TFAM protein expression. These findings provide insight into the efficacy of Jiawei Dachaihu decoction in the treatment of chronic stress-induced atherosclerosis, potentially through modulation of the SIRT1/PGC-1α/TFAM/Lon signaling pathway. The current study possesses certain limitations that warrant further investigation into the precise impact of SIRT1 on the interplay between Lon and TFAM proteins, in order to elucidate its significance in mitochondrial dysfunction. Additionally, forthcoming research endeavors should prioritize the elucidation of the specific role of TFAM protein in the protective efficacy of Jiawei Dachaihu decoction. This study still has some degree of limitations. For example, the interaction between Lon and TFAM was not experimentally verified. Moreover, the specific molecules acting on atherosclerosis were not investigated in depth to those in Jiawei Dachaihu. We will add to the exploration of the above two points in future studies. Nevertheless, our findings have provided valuable insights into the previously unexplored effects of Jiawei Dachaihu decoction and have indicated the potential targeting of the SIRT1/PGC-1α/TFAM/Lon signaling pathway for the treatment of chronic stress-induced atherosclerosis.

5. Conclusion

Jiawei Dachaihu Decoction efficiently alleviated chronic unpredictable mild stress in the Atherosclerosis animal model. Also, it exerted a protective effect against mitochondrial dysfunction by targeting the SIRT1/PGC-1α signaling pathway and the crosstalk with Lon protein. These findings showed that the SIRT1/PGC-1α/TFAM/Lon axis may be a potential therapeutic pathway in the treatment of AS-CUMS and it may maintain mitochondrial DNA stability.

Ethics approval and consent to participate

The study was conducted in accordance with the Declaration of Helsinki, and the protocol was approved by the Ethics Committee of Affiliated Hospital of Liaoning University of Traditional Chinese Medicine.

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CRediT authorship contribution statement

Haijiao Shi: Methodology, Data curation, Conceptualization. **Miao Sun:** Writing – review & editing, Writing – original draft, Investigation. **Shuai Wang:** Writing – review & editing, Writing – original draft, Formal analysis. **Fanyu He:** Writing – original draft, Resources, Methodology. **Ronglai Yang:** Validation, Project administration, Formal analysis. **Zheng Li:** Supervision, Methodology, Investigation. **Wei Chen:** Writing – review & editing, Writing – original draft, Data curation. **Fengrong Wang:** Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix ASupplementary data

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